Hyungseok Han

AI Infrastructure & Platform Engineer

dfafqfdaes159@gmail.com | GitHub | LinkedIn | Portfolio

I build AI infrastructure and developer platforms across cloud and on-premises environments. My work connects distributed-systems research with hands-on backend development, LLM inference, customer delivery, and production operations.

EXPERIENCE

Nota AI

Oct 2025 - Present

DevOps Engineer

Design and operate hybrid GPU infrastructure, reusable customer deployment architectures, and internal platforms for AI workloads.

NetsPresso / Arm AI Model Optimizer

- Designed and built Kubernetes service and deployment architecture for AWS and on-premises customers, standardizing Terraform and Helm/Kustomize configurations.
- Coordinated requirements, acceptance criteria, and two parallel customer deliveries; enabled final validation and packaging within two days after code freeze. The delivered system entered production as an early-access service in October 2026.
- Built customer observability and proposed trace-ID logging adopted by developers; support post-launch security response and operational stabilization.

GPU resource self-service platform

- Built a FastAPI/Next.js control plane for Proxmox GPU instance lifecycle, team quotas, and resource visibility; took a five-day PoC to internal production in approximately one month.
- Automated routine infrastructure support, recording three consecutive weeks without system-level support requests, excluding physical hardware failures.

LLM inference platform and KV-cache optimization

- Built inference platform foundations and evaluated KV-aware routing policies against model support and measured behavior; selected llm-d with approx-affinity routing.
- Contributed to LMCache offloading bottleneck analysis, performance patches, and request-cancellation crash investigation alongside other engineers.

EasyLLM - internal LLM gateway and serving platform

- Built a Python/FastAPI platform with Kubernetes, MLflow, and LiteLLM used by 70+ internal users; automated model mirroring, vLLM/llama.cpp serving, and benchmarks.
- Unified commercial and open-weight model access with per-user budgets and isolated serving paths for restricted workloads.

EXPERIENCE CONTINUED

MCP-based internal app and document platform | Nota AI

- Independently built and deployed a platform in October 2026 for permission-aware app discovery and document search by coding agents, using FastAPI, MCP, OpenFGA, and self-hosted hybrid retrieval.

goorm

Feb 2023 - Oct 2025

Infrastructure Engineer

Designed and operated multi-region production infrastructure for web IDE and code-judging platforms supporting thousands of concurrent code executions.

Reliability and developer productivity

- Built centralized logging across 10+ Kubernetes clusters in four regions, using Kafka to decouple collection, processing, and Elasticsearch/S3 storage.
- Redesigned container build and remote-cache workflows in Jenkins, reducing application build time by approximately 60% on average.
- Prototyped a Kubernetes code-judging architecture with Kafka-based execution/replay and storage failover; the architecture was later incorporated into the product.

Sunrise - LLM-based infrastructure operations agent

- Built a Python/AWS Bedrock agent with a Django interface for natural-language operations across thousands of cloud resources; adopted as a team support tool.
- Modeled inspection, execution, and error recovery as a state machine with AWS, Kubernetes, and Ansible domain agents; constrained execution to user credentials.

RESEARCH AND SPEAKING

ScaleOPT: A Scalable Optimal Page Replacement Policy Simulator

First author | ACM SIGMETRICS 2025 | Paper

Proposed a multicore-scalable simulator for optimal page replacement, connecting parallel data structures with systems performance evaluation.

Build with AI Seoul 2026 with Google DeepMind | Invited speaker, May 2026

Presented the GPU self-service platform from five-day PoC to internal production, including design-document and testing practices for AI-assisted development.

EDUCATION

Seoul National University | M.S., Computer Science and Engineering | 2021 - 2023

Distributed Computing Systems Lab; cache and distributed systems research.

Kwangwoon University | B.S., Software | 2017 - 2021

SELECTED SKILLS

AI systems: vLLM, llm-d, LMCache, LiteLLM, MLflow, MCP, retrieval systems

Platforms: Kubernetes, AWS, Proxmox/KVM, Terraform, Helm/Kustomize, Ansible

Development: Python, TypeScript/JavaScript, C/C++, FastAPI, Django, Next.js

Operations: Prometheus, Grafana, Loki, Elasticsearch, Kafka, Jenkins, GitOps